CECS 229 – Assignment 2

Base Functions

Given the file, 'pa2.py', complete the code for the following functions:

1. `b_rep(n, b)` – this function computes the base b-representation of a decimal integer, n, given base b.
    a. n – is a positive integer in decimal.
    b. b – is an integer between 2-36, representing the desired base.
    c. The returned value should be a string with the b-expansion of integer n.
2. `binary_add(a, b)` – this function calculates and returns the sum of two binary numbers, a and b.
    a. a – is a string made up of 0s and 1s with no spaces.
    b. b – is a string made up of 0s and 1s with no spaces.
    c. The returned value should be a string of 0s and 1s that is the result of adding the binary values, a and b.
3. `binary_mul(a, b)` – this function calculates and returns the product of two binary numbers, a and b.
    a. a – is a string made up of 0s and 1s with no spaces.
    b. b – is a string made up of 0s and 1s with no spaces.
    c. The returned value should be a string of 0s and 1s that is the result of multiplying the binary values, a and b.

Example Input/Output:

```
b_rep(10, 2)
1010
b_rep(10, 8)
12
b_rep(10, 16)
A
binary_add('101011', '11011')
1000110
binary_mul('101011', '11011')
10010001001
```

Notes:

1. Please place your name in the comment block at the top of the provided file.
2. Fill in the necessary code to complete the functions. You should not modify any of the provided code other than in the specified locations (ie. the TODOs).
3. Your solutions may not use any built-in functions that perform the equivalent task (ex. bin() or int(a, base=2)).
4. Test your code using the provided 'main.py' file. You can write additional tests in the 'your_tester.py' file to further check your solutions. Note: just because the main tests pass does not guarantee that it will pass the tests on Gradescope.
5. When you are finished, submit your 'pa2.py' file to Gradescope on Canvas for further testing and grading. It should allow you to make repeated submissions.